AILEIKE HART handheld device ALK475TREX modifies the damping time of the pressure transmitter

To improve the fluctuation interference of pressure measurement signals in industrial operating conditions and stabilize the control system's data acquisition, it is necessary to adjust the damping parameters of the pressure transmitter reasonably with the help of a HART handheld device. This article uses the HART handheld device ALK475TREX (Chinese version) as the operational carrier, and introduces two wiring implementation schemes: the built-in 24V power supply of the handheld device and the external power supply with a 250Ω load resistor. It demonstrates step-by-step the entire operation process of entering the parameter interface through the handheld device menu and modifying the transmitter damping time, providing standardized practical reference for on-site instrument operation and maintenance personnel.

1. Preliminary preparations

Prepare an ALK475TREX (Chinese version) handheld communicator, a pressure transmitter (HART protocol), and a communication cable for the handheld communicator.

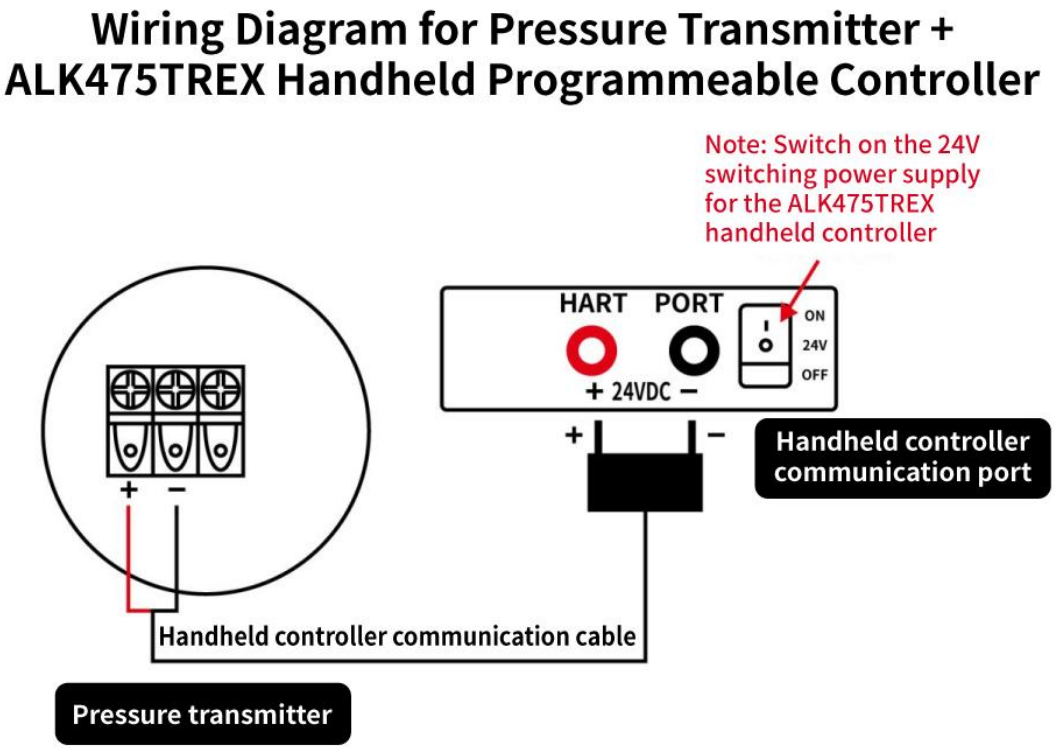
Tool connection

• Wiring method 1 (the ALK475TREX handheld communicator comes with a built-in 24V power supply, which can supply power to the transmitter):

(1) First, insert the communication cable connector into the communication interface of the handheld device (note: connect the positive pole of the communication cable connector to the positive pole of the handheld device's communication interface, and connect the negative pole to the negative pole). Then, turn on the 24V power supply.

(2) The other end of the communication cable is clamped onto the transmitter, with the red alligator clip attached to the positive terminal of the transmitter and the black alligator clip attached to the negative terminal of the transmitter.

(3) Complete the wiring, and ensure that the handheld device successfully supplies power to the transmitter.



Wiring diagram (I)

• Wiring method 2 (external power supply wiring for ALK475TREX handheld operator):

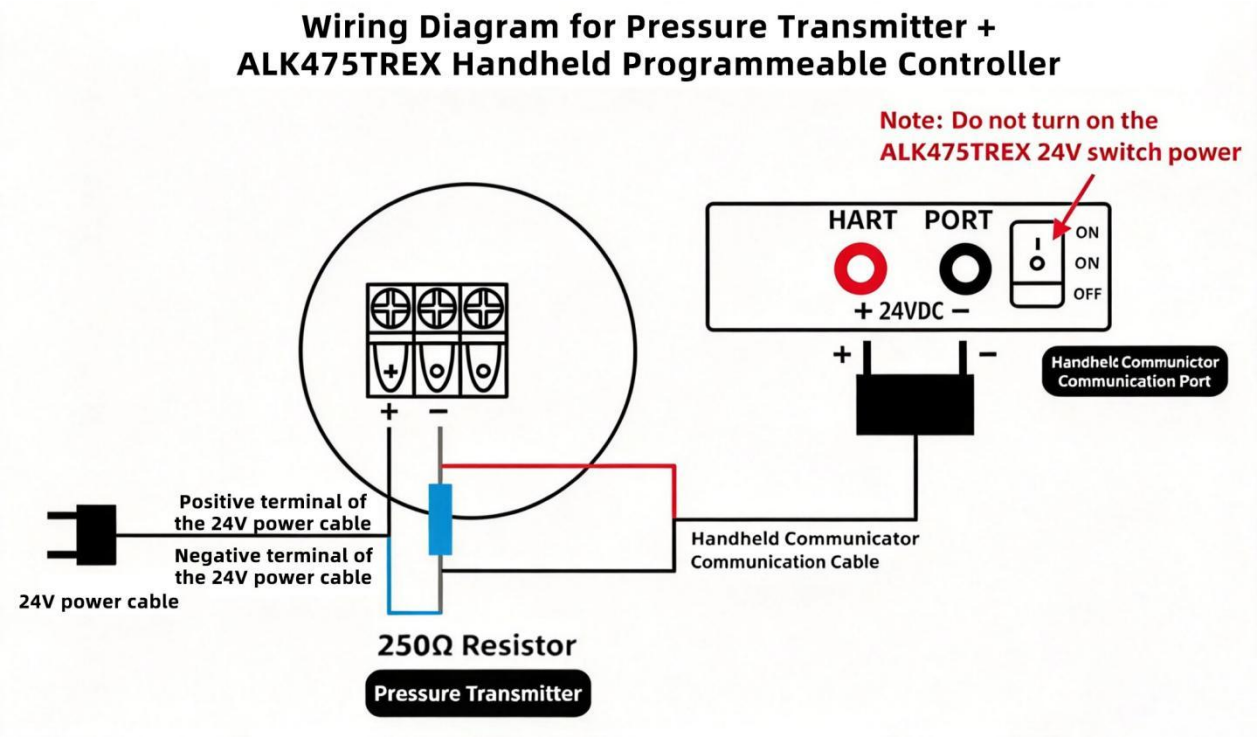
Additional preparations are required: an external 24V power cord and a 250Ω resistor

(1) First, connect the positive terminal of the external 24V power cord to the positive terminal of the transmitter power supply. Then, connect a 250Ω resistor to the negative terminal of the transmitter.

(2) Connect the negative terminal of the external 24V power cord to the other end of the 250Ω resistor.

(3) Plug the communication cable connector into the communication interface of the handheld device (Note: Connect the positive pole of the communication cable connector to the positive pole of the handheld device's communication interface, and connect the negative pole to the negative pole). Additionally, clamp the other two alligator clips on both sides of the 250Ω resistor.

(4) Finally, connect the external 24V power cord to the power supply. After all devices are successfully connected, take out the handheld device and turn it on. You can see that the handheld device has successfully communicated with the pressure transmitter.



Wiring diagram (II)

II. Modifying the damping time of the pressure transmitter using a HART handheld communicator

Currently, the handheld device has successfully established communication with the pressure transmitter. The specific operation is as follows:

(1) Select the pressure transmitter (see Figure 1), and press the right button to enter the pressure transmitter interface

(2) Press the key to select Configuration and Testing (see Figure 2), and press the right key to enter the Configuration and Testing interface

(3) Press the key to select Basic Settings (see Figure 3), and press the right key to enter the Basic Settings page

(4) Immediately press the key to select damping (see Figure 4), and then press the right key to enter the damping page

(5) Then press the middle button to clear the original damping time, and then enter the desired damping time (see Figure 5)

(6) Finally, press the right button to confirm, and the damping time of the transmitter is successfully modified (see Figure 6)

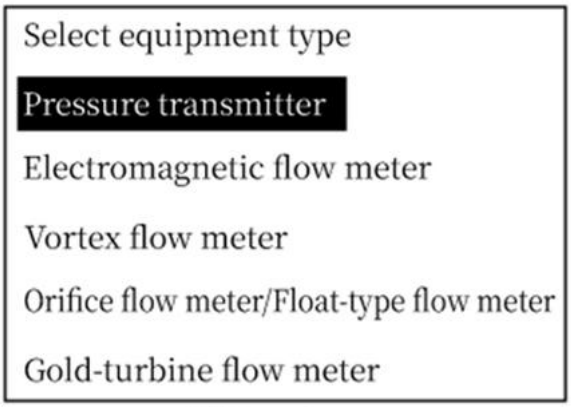


Figure 1

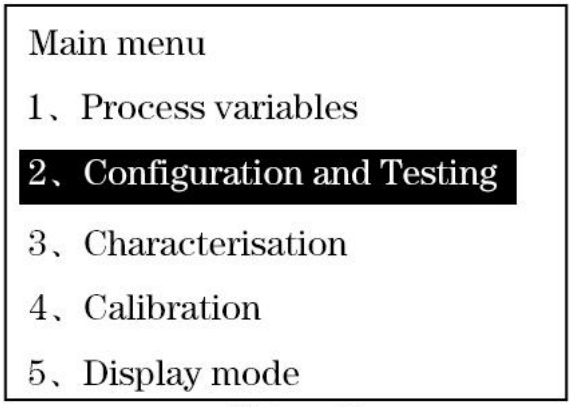


Figure 2

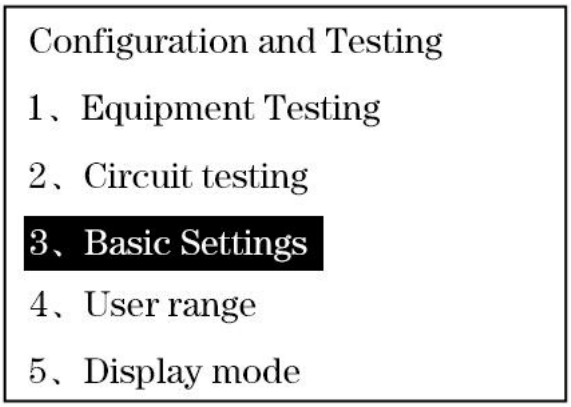


Figure 3

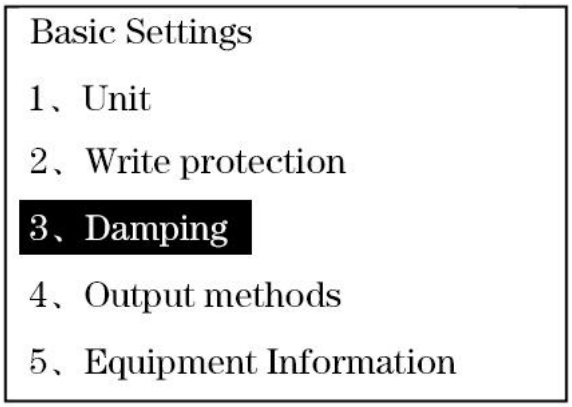


Figure 4

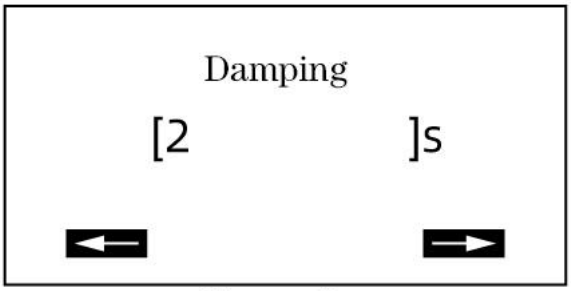


Figure 5

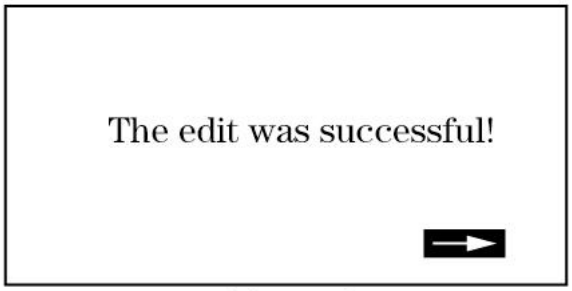


Figure 6